

PROJECT DESIGN PHASE

Electricity Consumption Analysis

Problem Solution Fit

Customer Segment

- Electricity Board Authorities
- Energy Analysts
- Government Organizations
- Utility Companies
- Researchers and Students
- Business Intelligence Professionals

Customer Constraints

- Large volume of electricity consumption data.
- Difficulty identifying usage trends manually.
- Lack of interactive visualization tools.
- Time-consuming report generation.
- Challenges in comparing state-wise and region-wise electricity consumption.

Available Solutions

- Microsoft Excel Reports
- Manual Data Analysis
- SQL Reports
- Static Graphs
- Traditional Reporting Systems

Jobs To Be Done

- Analyze electricity consumption efficiently.
- Compare electricity usage across different states.
- Identify high electricity-consuming regions.
- Monitor monthly electricity consumption trends.
- Visualize electricity usage on maps.
- Support data-driven decision-making.

Proposed Solution

Develop an interactive Electricity Consumption Analysis System using **MySQL, Tableau Desktop, Tableau Public, and Flask**.

The system stores electricity consumption data in a MySQL database, performs SQL-based analysis, and generates interactive dashboards using Tableau Desktop. The dashboard includes state-wise usage, region-wise usage, monthly electricity consumption trends, geographical maps, and top electricity-consuming states. The completed dashboard is published on Tableau Public and embedded into a Flask web application, allowing users to access the visualizations through a web browser.

Novelty

- Interactive Tableau dashboards.
- SQL-based data analysis.
- Geographic visualization using maps.
- Monthly trend analysis.
- Integration of Tableau Public with Flask.
- Web-based dashboard accessibility.
- Business insights through story visualization.

Social Impact

- Supports efficient energy planning.
- Helps understand electricity consumption patterns.
- Encourages data-driven decision-making.
- Improves electricity resource management.
- Assists government and utility organizations in policy planning.

Revenue Model

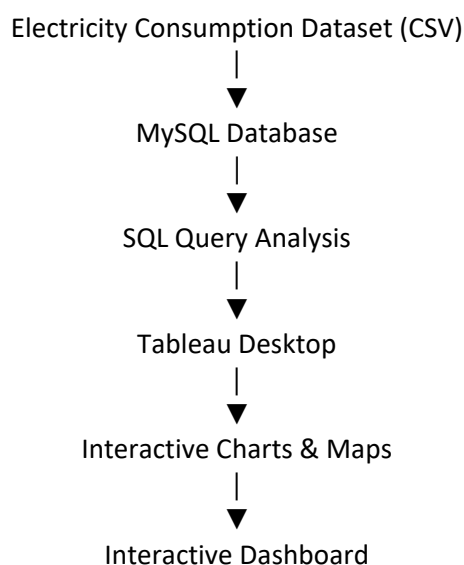
The project can be adopted by electricity boards, government agencies, and utility companies to improve operational efficiency and optimize electricity distribution. Better analysis of electricity consumption helps reduce resource wastage and supports cost-effective energy management.

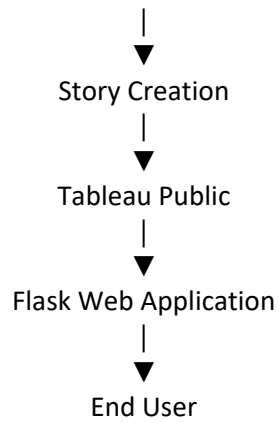
Scalability

The proposed solution can be extended to:

- National-level electricity monitoring.
- Real-time smart grid systems.
- Renewable energy monitoring.
- IoT-enabled electricity management.
- Predictive electricity demand forecasting.
- Multi-year electricity consumption analysis.

Solution Architecture





Technology Stack

Component	Technology Used
Dataset	CSV File
Database	MySQL
SQL Tool	MySQL Workbench
Data Visualization	Tableau Desktop
Dashboard Publishing	Tableau Public
Programming Language	Python
Web Framework	Flask
IDE	Visual Studio Code
Version Control	GitHub
Documentation	Microsoft Word & PDF

Advantages of the Proposed Design

- Stores data securely using MySQL.
- Performs efficient SQL-based analysis.
- Provides interactive visualizations.
- Displays geographical insights through maps.
- Offers an intuitive and user-friendly dashboard.
- Accessible online through Tableau Public.
- Integrated with Flask for web deployment.
- Easy to maintain and extend for future enhancements.